



Putative impacts of phytogetic additives to ameliorate lead toxicity in animal feed

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Abstract

Lead (Pb) is a toxic heavy metal and an environmental pollutant, particularly because of its anthropogenic activity. The main impacts of Pb is recognized to cause injurious influences of various levels of the tropic chain, due to bio-accumulated lead causes many health issues such as intoxication of different body organs, such as kidneys and liver, and reproductive and nervous systems. Industrial lead toxicity has reduced as a result of the attempts to decrease the lead levels in the surrounding work environment. Conversely, health risks related with long-term environmental exposure to a low dose of Pb have been steadily demonstrated. Long-term exposure to lead toxicity caused inflammatory infiltration, degenerative changes in testicular tissues, reduction in spermatocytes, necrosis of hepatocytes, degeneration in renal tubules, and renal epithelium hypertrophy. Hence, we need an influential approach to vanquish lead toxicity. This consequence has emerged the necessity for potentially safe represent remedy, favorably keeping both enhancement and chelating of the antioxidant competences. Many antioxidants have been used for chelating heavy toxic pollutants such as lead and oxidative stress released in excess during lead exposure. Several studies have stated the noticeable gathering of herbal singly or in combination in modulating lead-induced disturbances, therefore proposing great promise in enhancing health status and welfare of man as well as animals. For this, in the current review, we tried to discuss the enormous harmful influences of lead toxicity on the animal model and the disturbing truth that this detrimental toxic substance can be found quite simply in the surroundings and amplitude.

Keywords Lead toxicity · Oxidative stress · Herbal extract · Antioxidant

Introduction

The gathering of extreme levels of heavy metals tempts countless toxicity symptoms; thus, it is considered one of the oldest environmental issues and stays a critical health concern today

(Zhai et al. 2015). Poisonous heavy metals, counting lead, pose a serious threat to animal and human health (Abou-Kassem et al. 2016; Khafaga 2017; Saad et al. 2018). Both animals and human are exposed to Pb through drinking water, ambient air, food, electronic waste, industrial materials,

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contaminated soil, former industrial sites, and consumer products (WHO 2010; Nordberg et al. 2011). Today, developing countries are fronting the furthestmost serious Pb pollution problems, because of the increment of the industrial activities in these countries. Mainly, in human investigations, it has been well-known that children are very sensitive for Pb toxicity, because the human brain has slight capability for healing these harmful influences that are irreversible and untreatable which have been induced by Pb toxicity (WHO 2010; Hassanien and El Shahawy 2011; Bellinger et al. 2018). As a consequence of pollution, they cause declining in brain assignments and decrease in fulfillment that last during life (WHO 2010). In Egypt and China, it was observed that children had blood Pb level (BLL) above 10 µg/dL (Mostafa et al. 2009; Zhang et al. 2009). Likewise, as informed in 2010, the average BLL was 15.11 ± 5.62 µg/dL for children from Indian polluted village (Hegde et al. 2010). Based on the previous investigations in different regions of the world, exposure to heavy metal such as Pb causes the expansion of counteractive health impacts in humans and animals. Moreover, according to the animal studies, it has been indicated that Pb exposure induces neurologic and hematological dysfunctions (Lidsky and Schneider 2003), renal and hepatic damage (Sandhir and Gill 1995), and reproductive disorders (Ronis et al. 1996) in rats.

Several attempts have been used for attenuation of the adverse impacts of heavy metal poisoning. The most widely applied therapeutic approach to the toxicity of heavy metal is chelation remedy to encourage metal disposal from the body to the environment. Nevertheless, Pb poisoning chelators are themselves conveyed to have countless of various integrity and efficiency worries. So far, all the chelation treatments for Pb poisoning have not been permitted for clinical routine (McCarty 2012). The development of efficient and safe approaches versus Pb toxicity is hence a space of ongoing research. Lately, agricultural products, herbs and their extracts, have become one of the backbones of complementary and alternative remedies (Khafaga and Bayad 2016a, b; Abd El-Hack et al. 2015, 2018a, b; Alagawany et al. 2017; Dhama et al. 2018; Shewita and Taha 2018; Abd El-Moneim and Sabic 2019; Khafaga et al. 2019), which have displayed abundant properties against heavy metal toxicity (Yadav and Saxena 2017; Ewere et al. 2018; Waman et al. 2018). Thus, the attitudes of remediation and inhibition of Pb poisoning have been centralized on the use of intact herbs and herbal extracts (Alagawany et al. 2018). Phytogenic are capable to restrict the heavy metals and decrease their absorption through the small intestine and facilitate their excretion especially in the case of excessive poisoning through upholding to release the excretion mechanisms of the body. Herbs as feed ingredients can be added simply and affordably to the daily diet and can overcome the chelation therapy side effects. Herein, we review the prospective dietary approaches for Pb toxicological

effects on human, animals, and poultry health, and the potential modulatory role of herbal plants and their extracts against these negative effects.

What is lead?

Pb is one of the common toxic heavy metal in the environment. It is a bluish-gray color, easily shaped and molded, and has melting point; further, it can be mutual with other metals to form alloys. Pb is located on group 14 in the periodic table with atomic number and weight 82, 207.2, respectively, with chemical symbol Pb. It has 11.34 g/cm³ of density, and a low melting point of 621.43 °F. Lead has two different forms, organic and inorganic Pb is the most common. The inorganic form of Pb can be discovered in dust, soil, old paint, and different user products, while Tetra-ethyl Pb (organic Pb) is the lead form hired in leaded gasoline. Organic Pb complexes are excessively unsafe in biological systems, as they are absorbed via the skin, and are vastly toxic to the respiratory, renal, and nervous systems, far more than inorganic lead (WHO 2010).

Signs and symptoms of lead exposure

Presently, the exposure to any heavy metals has become a renowned source of disease worldwide. Among toxic elements reaching dangerous concentration are heavy metals; uranium, mercury, arsenic, nickel, cadmium, selenium, chromium, zinc, silver, and gold. The hazard of heavy metals is aggravated by their almost indefinite perseverance in the environment due to their steady form in the nature. Lead toxicity is considered one of the utmost widespread and best comprehend infancy disorders of poisonous ecological source. It has been indicated that Pb has no critical function in the body of animals and human, and Pb toxicity accounts for about 0.6% of the total problems of disease (WHO 2009). The main sources of human or animal exposure to Pb are systems of drinking water with pipes and lead solder, ceramic glazes, Pb in electronic waste (e-waste), Pb in the food chain and in solder of food cans, lead released by incineration of Pb-containing waste via contaminated soil, Pb from active industries (such as mining), and Pb pollution as an inheritance of historical pollution from former manufacturing regions (WHO 2010). In children, there are some apparent signs after Pb toxicity such as loss of appetite, headaches, hyperactivity, fatigue, anemia, metallic taste, decreased nerve conduction, behavioral problems, weight loss, insomnia, delayed mental growth and learning, and possibly motor neuron disorders (Bellinger et al. 2018). Exposed children to Pb toxicity in the early life are suffering from limitation of attention span and disturbance of behavior, since the cerebrum of *Homo*

sapiens has little capability for reparation and these impacts are irreversible and untreatable. They cause decrease in brain assignments and physiological function and lessening in achievement that last throughout life (WHO 2010; Bellinger et al. 2018). Lead, in its natural form, is a non-reactive metallic element that tempts a wide range of physiological, biochemical, behavioral, and neurological disorders in both animals and humans (Suradkar et al. 2009). Lead can enter the body absorbed by the skin, ingested or inhaled, and is then circulated in the blood stream and translocated to soft organs such as brain, liver, lungs, and kidney and mineralized tissues (teeth and bones). The consequence toxicity is predicated centered on the interactions of indispensable metallic elements with Pb (Ahamed and Siddiqui 2007; Vesey 2010), and oxidative stress caused by the stripping of the binding sites that are fundamentally obligatory for the body minerals (Farmand et al. 2005; Liu et al. 2009). This retrogradation resistance in biologic systems and, predisposes, hypothetically it to either secondary or opportunistic infections. In several animal studies, broilers exposed to high levels of Pb acetate (200 mg/kg) in the diet exhibited countless of clinical sign: anorexia, regurgitation, weight loss, prostration, wing droop, leg paresis, lethargy, greenish diarrhea, and gross change (congestion of kidney, spleen and liver, necrotic foci, gizzard lining, hemorrhages on muscles) (Jaiswal et al. 2017). Moreover, in another study which was carried out by Rahman and Joshi (2009), it was concluded that the impacts of Pb acetate (250–400 ppm in drinking water) led to depression in feed intake, body weight gain, and diminishing growth indices in broilers. It has been suggested that the main pathways of Pb-persuaded poisoning is interrelated to highly generation of oxidative stress (Fig. 1). Pb-induced oxidative stress is connected with two various pathways; the first promotes the synthesis of ROS such as hydroperoxides (HO₂) and oxygen hydrogen peroxide (H₂O₂), and the next reduces the antioxidant reserve activities and becomes exhausted (Flora et al. 2012). Another work verified that oxidative stress induced by Pb could cause lipid peroxidation in erythrocytes membranes and, as a consequence, results in erythrocytes burst and hemolytic anemia (Flora 2002).

Most animal experiments conveyed that Pb could transport in the body via blood circulation and accumulate in soft tissues, bones, and other pivotal organs (Gangoso et al. 2009). Typically, the bones include about 90% of the whole amount of Pb in the body, and it is (excessive amounts of Pb) mostly by replacing calcium. Hence, a decrease in bone mineral density (BMD) can occur as a result of exposure to Pb (Theppeang et al. 2008). Regarding the impacts of Pb on gene expression, Pb could alter the expression of selenoprotein related genes in the cartilage tissue of broiler (Gao et al. 2016). Likewise, Pb minimizes the antioxidant defense enzymes such as catalase, superoxide dismutase, and glutathione and changes activities and expression of antioxidant enzyme-related

genes (Kasperczyk et al. 2012, 2013). Unfortunately, the massive cost attendant with the removal of pollutants from water, feeds, soils, and industrial sites by different ways of traditional physiochemical approaches has been promising companies to ignore the problem

The biological role of Pb in animals

For many years, it was thought that Pb is not required in animals or human nutrition. Nevertheless, trace quantities of Pb (29 ng/g diet) have been described to be essential for several cellular systems and enzyme activities all over the body; in rats, Pb may play an important role during cell development, reproduction, and hematopoiesis (Assi et al. 2016). Humans are exposed to salts and oxides of Pb meanwhile numerous environmental sources, such as paint, atmospheric dust, automobile exhaust, or polluted food and water.

Previously and till the food canning industry has transformed from Pb-soldered to Pb-free cans, the foremost supplier was the Pb that leached into canned foods (Chisolm Jr 1985). At current, the main sources of lead exposure are probable to be dust particles from the soil transmitted into homes, paints, or drinking water.

The Food and Agriculture Organization and World Health Organization (WHO 1978) mentioned that acceptable concentration for adults of daily Pb intake up to 7 µg/kg body weight or 490 µg of Pb. There is no similar standard for children and infants. However, Ziegler et al. (1978) found in metabolic balance investigation that the daily Pb consumption exceeded 5 µg/kg per day was accompanied with positive Pb balance in children up to 2 years of age.

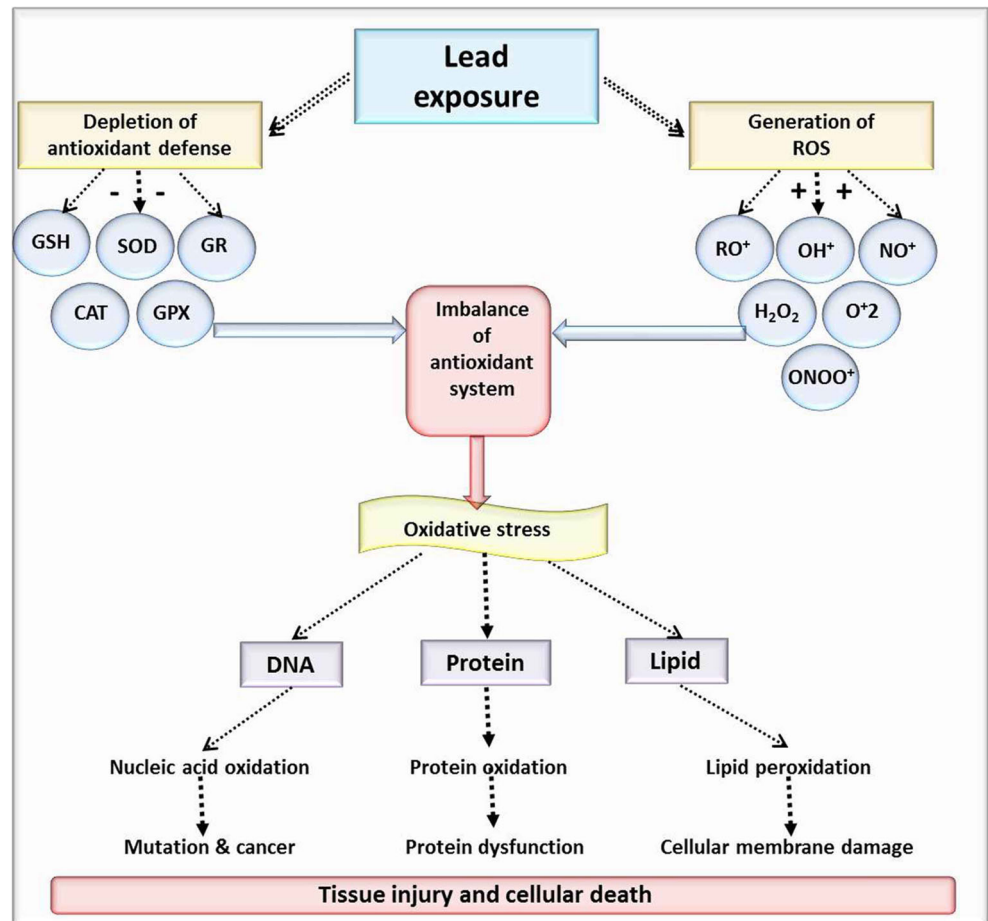
Long-term exposure of adults to Pb can result in impaired cognitive performance and nervous system functions. However, infants and young children are more sensitive to even low levels of lead, which may contribute to behavioral problems, learning deficits and lowered IQ (Rubin and Strayer 2008; Wani et al. 2015)

Morphological and behavioral pattern for lead toxicity in animal

Symptoms apparent of Pb toxicity have been well-documented by various experimental researchers; however, Pb poisoning symptoms certainly take a diversity of shapes according to several factors such as the exposure period, species of animals, dose of Pb toxicity, the circumstances, and individual appearances, and slight signs may appear, or there may even be no noticeable signs whatsoever on some cases.

Generally, at low exposure level, Pb has acute and chronic effects on the live animal's health. Pb exposure at higher levels has massive range of deleterious influences on many organs in

Fig. 1 Pathways of Pb-persuaded poisoning



the body of animals, including reduced feed consumption and growth rate (Taha et al. 2019). Moreover, body weight, function of ovary, and egg production were diminished by exposing Pb level in laying Japanese quail diets (Alagawany et al. 2018; Farag et al. 2018); further, hematobiochemical alterations were induced by Pb toxicity in quails (Humayun et al. 2015). It has been mentioned that the degree of signs which appeared on animals depends on the form of Pb where organic Pb poisoning is more stronger than inorganic Pb because it has a higher lipid solubility, so its symptoms are mostly connected to the central nervous system, such as cognition disorders, insomnia, and delirium, as itemized by Mustafa and Hussein (2015); rats exposure to Pb acetate exhibited numerous undesirable neurological symptoms, however, such as a reduction in vigor, tremor; weakness of muscle-mass; along with instability and imbalance; an irregular gait. One precarious occurrence during the differentiation of astrocyte is the expression alterations of the glial marker alleged glial fibrillary acidic protein (Feeser and Loria 2011). Pb acetate can be found deposited in the cerebellum, disturbing its physiology as well as causing tremors (Hsu and Guo 2002), cellular deterioration, neurotoxicity, and possibly cellular apoptosis (Mtui et al. 2011). Regarding behavioral changes which are subordinate

to Pb toxicity, exposure are similar in human, rodents, and primates. A study revealed by Rice (1992) indicated that monkeys fed Pb for 6.67 months have gotten blood Pb concentrations from 3 to 25 g/dl, and in advances ages, it was observed that monkeys treated with Pb displayed less ability to learn, mostly at longer delay intervals (Rice 1992). Bird's clinical symptoms of Pb-induced toxicity include anorexia, wing droop, anemia, and neurological disorders (Franson and Pain 2011). Experimental, sublethal Pb level of toxicity in chicks influenced balance, locomotion, and depth perception, as well increase risk of collision (Burger and Gochfeld 2000). The formed Pb salts enter the gastrointestinal tract, where they are absorbed in the intestine and transported by the blood-stream, subsequently reaching the bird's tissues and organs. Few days later, the toxicity symptoms start to appear (Friend 1987). Abnormal behavioral signs in birds were observed such as anatomical malformations, landing accidents, and uncommon locations of the head and/or vocal or neck alterations (high-pitched honk as a result of Pb toxicity (Friend 1987; Sowden 1988). Moreover, the green cloacal feathers are a very typical sign for Pb toxicity due to intense green-colored diarrhea. After 2 weeks, there is a paralysis in the gastrointestinal tract stopping the digestion of any ingested feed. This makes

the birds become very weak and loosed their weight, with the following difficulty in walking or flying. The tips of wings also hang regularly on the ground or float in the water. In the other hand, several authors do not observe any direct link between plumbism and weight loss, but noticed that some birds with obvious signs of Pb toxicity have consumed feed a short time prior dying (Beer and Stanley 1965). Exposure to higher level of Pb led to cause neurotoxicity (peripheral neurons and the central nervous system) in different animal species and human, and it has been well studied. Moreover, the exposure to Pb has some signs where there is central nervous system (CNS) damage—they may cease grazing, walk aimlessly, appear dull and unresponsive, and/or be blind. In some cases, these symptoms are attended by paralysis of the tongue, muscle twitches, circling, and “stargazing.” Get immediate veterinary advice if your livestock show CNS signs; a rapid and accurate diagnosis is vital to preventing continuing losses. Taha et al. (2019) indicated that exposure of rabbits to Pb acetate resulted in adverse influences on cytological and molecular description of rabbits.

Histological pattern for Pb toxicity in animal

Histopathological lesions in liver comprised congestion, degeneration of hepatocytes and fatty alteration, and whereas in kidneys, casts in lumen cellular infiltration, tubular necrosis, and mild fibrosis in interstitial tissue in Japanese quails (Farg et al. 2018). Histopathological alterations in the renal proximal tubular epithelium were derived from plumbism, and these alterations caused interstitial nephritis and typically associated with hypertension. It's accumulation in proximal involuted tubules of renal cortex that generate both morphological and biochemical substantiation of Pb poisoning. An extensive work has been well-documented to investigate the severe impacts of Pb toxicity towards the kidneys (Nisar et al. 2014). Animals exposed to Pb toxicity showed degeneration, hemorrhages, and loss of striations in muscle. Liver and kidney are the most organs to have been affected by toxic heavy metals such as and exhibited cloudy swelling, degeneration, Fatty degeneration, hemorrhages, and serous exudates. Spleen showed hemorrhage, necrosis of red, white pulp, and degeneration of the lymphoid follicles induced by Pb exposure.

The application of medicinal plants to reduce Pb toxicity (Table 1)

Allium sativum (garlic)

Allium sativum has been used with effective outcomes in traditional medicine against several diseases. This effectiveness

may be due to its high amount of water-soluble organic sulfur compounds, lipid soluble compounds, and S-allylcysteine. This content may act as a prospective candidate in immune improving and antioxidant activities. It demonstrates phenomenal attenuating features versus toxic heavy metal owing to its ownership of chemicals comprising amino acids, carbohydrates, volatile oils, enzymes, and organosulfur groups. Garlic is best recognized for antioxidant, anticarcinogenic, antibacterial, antiallergic, immunostimulatory, and antianabolic impacts, as well as the influence of specific enzyme inhibitors (Mikaili et al. 2013; Schäfer and Kaschula 2014; Arreola et al. 2015).

Garlic was comprehensively contributed to attenuate the metal-induced poisoning. Recent supportive evidence indicates that garlic contains components capable of detoxifying Pb, cadmium, methyl mercury, phenyl mercury, and arsenic. The restorative properties of garlic have been attributed to its antioxidant activities and/or chelating efficacy (Flora and Pachauri 2010; Arreola et al. 2015). Dietary administration of garlic conspicuously reduced the clastogenic effects of heavy metals. Moreover, fatal effects of mitochondrial injury and oxidative stress induced by body metal burden were effectively reduced by garlic. Garlic was superior in its curative effect to those of D-penicillamine and 2,3-dimercapto-1-propanol, N-acetyl DL-penicillamine and 2,3-dimercaptosuccinic acid, and the current remedies. Many researchers have used garlic extract attenuates Pb-induced neural, renal, hepatic, and hematic toxicity in rats and reduce mitochondrial damage and apoptosis in tissue culture models (Sharma et al. 2010; Lawal and Ellis 2011). It was well recognized that broiler chickens are vulnerable to Pb toxicity. In light of this, it has been indicated that dietary Pb level of (1.0 mg/kg) can induce substantial reduction in the broiler growth performance and consistent depression in blood level of erythrocyte enzyme called d-aminolevulinic acid dehydratase (Bakalli et al. 1995). A considerable number of studies have shown that garlic administration is protective against Pb toxicity in chickens (Hanafy et al. 1994) and in rabbits (Shehata 2011). Feeding the broiler chickens on garlic (1, 2, or 4% *Allium sativum*) improved the alterations of growth performance except feed conversion rate caused by Pb toxicity (100 mg/kg) (Hossain et al. 2014a), and enhanced significantly the values of blood variables (Hossain et al. 2014b). Based on the previous studies, the protective properties of garlic against Pb toxicity can be attributed to (1) its chelation ability, provided by compounds with free amino and carboxyl groups and sulfur-containing amino acids, which in turn enhances the Pb excretion from the body; (2) its antioxidative role, provided by organo-sulfur components such as diallyl tetrasulfide; and (3) the prevention of intestinal absorption of Pb, by its amino acids containing sulfur group such as S-allyl mercaptocysteine and S-allyl cysteine.

Table 1 Major findings of previous investigations regarding the amelioration impacts of phytogetic plants against lead-induced toxicities

Herb	Common name	Active ingredients	Mechanisms of action	References
<i>Allium sativum</i>	Garlic	• High amount of water-soluble organic sulfur compounds • S-allylcysteine • Lipid soluble compounds • Volatile oils and enzymes	• Antioxidant • Anticarcinogenic • Antibacterial • Antiallergic • Immunostimulatory • Antianabolic • The influence of specific enzyme inhibitors • Antioxidant activities • Chelating ability • Prevention of intestinal absorption of lead, by its amino acids containing sulfur group such as S-allyl mercaptocysteine and S-allyl cysteine • Reduce mitochondrial damage • Reduce apoptosis in tissue culture models	Mikaili et al. (2013) Schäfer and Kaschula (2014) Arreola et al. (2015) Flora and Pachauri (2010) Arreola et al. (2015) Sharma et al. (2010) Lawal and Ellis (2011)
<i>Garcinia kola</i>	Heckel (bitter cola)	• Carbazole alkaloid • Phenolic compounds • Organic compounds • Inorganic compounds	• Masticatory substance to stimulate the saliva flow • Eaten as an aphrodisiac • Ameliorating a varied respiratory diseases; cough and asthma. • Antidote, snakebites, enhance fertility in males, and management of dysentery	Leakey (2001) Omode et al. (1995) Ajani et al. (2008) Daramola and Adegoke (2011)
<i>Garcinia kola</i>	Heckel (bitter cola)		• Antioxidant activity • Antihepatotoxic efficacy against carbon tetrachloride, hepatotoxicants, phalloidine, thioacetamide, galactosamine, and ethanol in animals • Attenuate the chronic lead acetate poisoning	Daramola and Adegoke (2011) Adaramoye et al. (2005a, b) Osemwegie et al. (2017)
<i>Coriandrum sativum</i>	Coriander	• Coriandrin • Isocoumarines	• Effective antioxidant • Treating stomach ulcers and other abdominal problems • Enhance the oxidative status after intoxication with lead in male rats • Reduce oxidative stress in the kidney, maybe via reduced levels in heavy metals	Taniguchi et al. (1996) Sharma et al. (2010) Bhat et al. (2014) Eidi et al. (2009) Chithra and Leelamma (2000) Nishio et al. (2019)
<i>Yucca schidigera</i>	Yucca	• Resveratrol • Saponins • Several enzymes • Several antioxidants agents	• Hypocholesterolemic • Hypoglycemic • Antioxidant • Immunostimulatory • Anticarcinogenic • Antiinflammatory • Maintain ammonia emission and reduce its level in poultry farms • Decline blood urea contents	Chrenkova et al. (2012) Alagawany et al. (2016) Alagawany et al. (2018) Farag et al. (2018) Piacente et al. (2005)
<i>Yucca schidigera</i>	Yucca		• Great absorption capability for deleterious volatile compounds, such as hydrogen sulfide and ammonia • Potential modulatory function towards the Pb-induced inhibitory impacts on the productive and reproductive traits of Japanese quails • Enhanced immunohistochemical and histomorphometrical alterations caused by Pb • Decreasing levels of inflammatory indicators including nitric oxide (NO), vascular endothelial growth factor (VEGF), tumor necrosis factor alpha (TNF-α) and transforming growth factor-β1 (TGF-β1) • Regulates glucose homeostasis in co-exposed quails	Vlckova et al. (2017) Alagawany et al. (2018) Taha et al. (2019) Alagawany et al. (2018) Farag et al. (2018) Farag et al. (2018)

***Garcinia kola* Heckel (bitter cola)**

Garcinia kola (bitter kola) is a conventional widely utilized herb which is identified as a spice broadly in Nigeria and other countries in Africa. It is as African wonder nut belongs to *guttiferae* family and cultivates in coastal rainforests regions. Conventionally, *Garcinia kola* nuts are gummed as masticatory substance to stimulate the saliva flow (Leakey 2001). The nuts kernels are widely traded and eaten as an aphrodisiac (Omode et al. 1995; Leakey 2001). *Garcinia kola* is also greatly valued since its medical profits (Hertog et al. 1993). In African medicine, it possesses ameliorating a varied respiratory diseases; cough and asthma. Moreover, it has been used as antidote, snakebites, enhance fertility in males, and management of dysentery (Ajani et al. 2008; Daramola and Adegoke 2011). Currently, it commences to be hired in toxicological investigation as a therapy for several toxic heavy metals and drugs due to its content of carbazole alkaloid, phenolic, organic, and inorganic compounds which are responsible for its antioxidant activities (Daramola and Adegoke 2011). Extensive evidence has been presented in support of the antihepatotoxic efficacy of *Garcinia kola* in animals against such carbon tetrachloride, hepatotoxicants, phalloidine, thioacetamide, galactosamine, and ethanol (Adaramoye et al. 2005a, b). Recently, Osemwegie et al. (2017) assessed the efficiency of *Garcinia kola* aqueous extracts to attenuate the chronic Pb acetate poisoning in Wistar rats as experimental models.

***Coriandrum sativum* (coriander)**

Coriander (*Coriandrum sativum*), is a common species of “Umbelliferae” plant family; it has several beneficial medicinal activities. Many experimental studies stated the medicinal and nutritional function of *Coriandrum sativum* supplements in various species of animals (Eidi et al. 2009; Sharma et al. 2010; Bhat et al. 2014). Coriandrin and isocoumarines represent the main constituents of this herb and this elucidates why this herb is utilized as an effective antioxidant (Taniguchi et al. 1996). In alternative medicine field, coriander has a tremendous value for the treatment of stomach ulcers and other abdominal problems. In the same context, it was reported that *Coriandrum sativum* has a potential antioxidant activities with enhancement of oxidative status after intoxication with Pb in male rats (Sharma et al. 2010). In addition, treatment with the hydroalcoholic seed extract of *C. sativum* resulted in a tissue-specific amelioration of oxidative stress produced by lead (Velaga et al. 2014). While treatment with methanolic extract of coriander led to decreased concentration of Pb and attenuation of hisopathologic liver damages of rats tissues (Téllez-López et al. 2017). Recently, Nishio et al. (2019) confirmed that coriander leaf extract intake participates to powerful

resistance to oxidative stress in the kidney, maybe via reduced levels in heavy metals.

Yucca schidigera

Yucca schidigera is a Mexican medicinal plant which is utilized efficiently and conventionally to improve the general health condition. It has several favorable properties such as growth promoter, hypocholesterolemic, hypoglycemic, antioxidant, immunostimulatory, anticarcinogenic, and antiinflammatory (Alagawany et al. 2016; Saeed et al. 2018). Additionally, it is a commercial origin of resveratrol, saponins, several enzymes, and antioxidants agents (Chrenkova et al. 2012; Alagawany et al. 2018; Farag et al. 2018). It is also recognized for beverages and cosmetics industry, and as supplement for animal and poultry diets. Likewise, *Yucca* can maintain ammonia emission and reduce its level in poultry farms as well as decline blood urea contents (Piacente et al. 2005). *Yucca* also has a great absorption capability for deleterious volatile compounds, such as hydrogen sulfide and ammonia (Vlckova et al. 2017). Recently, Alagawany et al. (2018) revealed that *Yucca* exhibited a potential modulatory function towards the Pb-induced inhibitory impacts on the productive and reproductive traits of Japanese quails. The authors revealed that the treatment with *Yucca schidigera* especially at level of 200 mg/kg diet was effective in reversing the Pb-induced alterations as well as enhanced immunohistochemical and histomorphometrical alterations caused by Pb, and these favorable impacts could be attributed to its antioxidant ability (Alagawany et al. 2018; Farag et al. 2018; Taha et al. 2019) and decreasing levels of inflammatory indicators including nitric oxide (NO), vascular endothelial growth factor (VEGF), tumor necrosis factor alpha (TNF- α), and transforming growth factor- β 1 (TGF- β 1). Besides, *Yucca schidigera* extract has been found to be able to regulate glucose homeostasis significantly in co-exposed quails (Farag et al. 2018).

Conclusions

Numerous investigations have studied the occupational exposure and health effects of Pb, and numerous efforts to avoid Pb-related health problems have resulted in a significant reduction in the number of occupational Pb toxicity cases. As attention in lead's carcinogenicity and the health disorders of environmental lead exposure has grown, we scrutinized in this review the health and toxicity effects of Pb, and reviewed the latest literatures on the observations and treatments of lead exposure. The exposure to elevated concentrations of Pb has undesirable impacts on poultry and animal, comprising reducing growth performance and immunohistochemical and histomorphometrical alterations of the internal organs in the

body. Numerous herbal plants and their extracts and active compounds can possess immunomodulation, antioxidant, and other favorable health influences which can create them an influential approach in avoiding or decreasing the deleterious effects of Pb. It is recommended to include these herbal plants or their products in animal and human diets to fight the possible exposure to Pb toxicity.

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